TEACH YOURSELF INTERMEDIATE XI UNIT 3

Programming Fundamentals (Python)

IDE, variables, control flow, and libraries

What you will master

- Explain low/mid/high-level languages and why Python is beginner-friendly
- Use an IDE (e.g. VS Code) to write and debug programs
- Apply selection and repetition in Python
- Import built-in and third-party libraries

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

3.1 Programming & language levels

A **program** is a set of instructions. Languages sit on a spectrum:

- **Low-level** — machine/binary (fast, hard to read)
- **Mid-level** — e.g. assembly (mnemonics)
- **High-level** — Python, Java (readable, portable)

Python emphasizes readability — ideal for learning logic, data work, web (Django/Flask), and automation.

3.2 Python building blocks

```
# variables & types
name = "Ali"
age = 16
average = 78.5
is_enrolled = True

# selection
if average >= 50:
    print("Pass")
else:
    print("Improve")

# repetition
for i in range(1, 6):
    print("*" * i)
```

Common types: `int`, `float`, `str`, `bool`. Indentation defines blocks — no braces.

3.3 IDE workflow

VS Code + Python extension gives editing, syntax highlight, run, and debugger. Workflow: edit → run → read traceback → fix → repeat.

Try it

Write a program that asks for two numbers and prints sum, difference, product, and quotient (guard divide-by-zero).

3.4 Libraries

```
import math
print(math.sqrt(144))

# third-party (install via pip)
# pip install requests
```

Libraries reuse tested code — you focus on problem logic instead of reinventing wheels.

Teach Yourself — Practice & Check

1. Print multiplication table of 7 using a loop.
2. Convert Celsius to Fahrenheit: — .
3. Debug: why does `if x = 5` fail in Python?